Muhammad Umair

AI/ML Engineer | Backend-Trained, ML-Focused Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer transitioning from backend/full-stack development into AI/ML Engineering, combining a strong foundation in scalable API design, database architecture, and production deployment with hands-on machine learning and NLP skills (PyTorch, scikit-learn, pandas, NumPy, Hugging Face, LangChain). Experienced building and shipping ML-powered features end-to-end — from model/data pipeline to a production backend that serves them — including a regression-based prediction model and an AI-driven pose-detection microservice. Comfortable operating across the full stack: data preparation, model integration, API design, and deployment.

TECHNICAL SKILLS

Machine Learning: PyTorch, scikit-learn, pandas, NumPy, regression modeling, feature engineering, model evaluation

NLP & LLMs: Hugging Face, LangChain, LLM integration, prompt engineering

AI Systems Integration: AI/ML microservices with FastAPI, real-time inference pipelines, pose-detection & CV-driven features

Languages: Python, JavaScript, C++

Backend: Node.js, Express.js, NestJS, Django, FastAPI, REST APIs, Microservices

Frontend: React.js, HTML, CSS

Databases: PostgreSQL, MongoDB, MySQL, Redis

DevOps / MLOps: AWS, Docker (Basic), GitHub Actions, CI/CD, Vercel

Security: JWT Authentication, RBAC, API Security

Tools: Git, Postman, Jupyter, Stripe API, MongoDB Compass

MACHINE LEARNING & AI PROJECTS

Vehicle Price Prediction Engine *Python • scikit-learn • pandas • NumPy*

- Trained and evaluated a regression-based machine learning model to predict vehicle prices from real-world data.
- Achieved 85% prediction accuracy through rigorous feature engineering, data cleaning, and model tuning.
- Ran the full ML workflow end-to-end: data preprocessing, feature selection, model training/evaluation, and result analysis.

AI-Powered Fitness Tracking System *Django • FastAPI • PostgreSQL • AI Pose Detection*

- Architected a scalable backend using Django and PostgreSQL to manage high-volume fitness and workout data.
- Integrated FastAPI microservices to power AI-based pose detection and real-time exercise analysis, connecting model inference to production APIs.
- Engineered a modular tracking system enabling seamless real-time monitoring of user workouts, with AI processing decoupled from core backend services via clean architecture.

SOFTWARE ENGINEERING EXPERIENCE

Backend Developer — Proder Cave *Sep 2025 – Dec 2025*

Quran UI Project Repository

- Engineered robust backend services for the Quran UI platform utilizing Node.js.
- Designed and deployed scalable RESTful APIs to efficiently manage structured data and metadata.
- Implemented clean architecture principles, significantly improving code modularity and long-term maintainability.
- Optimized complex database queries, resulting in enhanced API response times and overall system performance.
- Partnered with frontend developers to ensure seamless system integration and data flow.

ADDITIONAL ENGINEERING PROJECTS

Professional Networking & Job Board Platform *React • Node.js • Express • MongoDB*

- Engineered a full-stack networking platform featuring robust authentication and user profile management.
- Designed and implemented scalable RESTful APIs utilizing MongoDB and Mongoose.
- Developed a dynamic, responsive frontend using React, Context API, and Axios.

Scalable Hospitality Operations Platform *NestJS • PostgreSQL • REST*

- Architected a modular backend system tailored for complex booking and operational workflows.

- Integrated secure authentication protocols and Role-Based Access Control (RBAC).
- Streamlined and optimized PostgreSQL databases, boosting API performance and data retrieval speed.

Real-Time Ticketing & Reservation System *MERN • NestJS • MongoDB*

- Developed a full-stack booking application supporting real-time seat reservations and secure payment processing.
- Implemented secure JWT-based authentication and fine-tuned MongoDB queries for optimal efficiency.

EDUCATION

Bachelor of Software Engineering — Information Technology University (ITU), Lahore, Pakistan

CORE STRENGTHS

Machine Learning & Model Integration • Data Preprocessing & Feature Engineering • Backend Architecture • API Design • Authentication & Security • Database Optimization • CI/CD Deployment • Clean Code • Problem Solving